Measurements and Units

Quantities in the laboratory carry units, and many of those units have powers. At sea level the density of air is about 1.2 kg $\mathrm{m^{-3}}$, and a barometer reading corresponds to a pressure of 1.1 times 10^5 $\mathrm{N/m^2}$. The solution contains 0.1 mol $\mathrm{dm^{-3}}$ of hydrochloric acid, so 250 $\mathrm{cm^3}$ of it holds 0.025 mol. The reservoir holds 2.5 $\mathrm{km^3}$ of water spread over 150 $\mathrm{km^2}$. Near the surface of the earth, free fall has an acceleration of 9.81 m $\mathrm{s^{-2}}$.

The intensity of sunlight at the ground is roughly 950 W $\mathrm{m^{-2}}$ on a clear day. The density of ice is 0.92 $\mathrm{g/cm^3}$, so a block of volume 60 $\mathrm{cm^3}$ has a mass of 55 g. The pump delivers 0.8 $\mathrm{m^3/s}$, which is 800 $\mathrm{dm^3}$ every second. The specific heat capacity of aluminium is 0.9 J $\mathrm{g^{-1}}$ $\mathrm{K^{-1}}$.

The plate has an area of 0.5 $\mathrm{m^2}$ and a thickness of 10 cm, so its volume is 0.05 $\mathrm{m^3}$. A steel wire with a cross-section of 10.0 $\mathrm{mm^2}$ carries a load of 900 N, which is a stress of 90 $\mathrm{N/mm^2}$. The measured viscosity of the ice melt is 0.5 kg $\mathrm{m^{-1}}$ $\mathrm{s^{-1}}$. A cart accelerates at 4.5 $\mathrm{m/s^2}$ for 6 s and reaches a speed of 27 m/s.

The plate has an area of 2.0 $\mathrm{m^2}$ and a thickness of 2 cm, so its volume is 0.04 $\mathrm{m^3}$. The pump delivers 0.4 $\mathrm{m^3/s}$, which is 400 $\mathrm{dm^3}$ every second. The measured viscosity of the aluminium melt is 0.5 kg $\mathrm{m^{-1}}$ $\mathrm{s^{-1}}$. The specific heat capacity of aluminium is 0.9 J $\mathrm{g^{-1}}$ $\mathrm{K^{-1}}$.

A cart accelerates at 3.0 $\mathrm{m/s^2}$ for 4 s and reaches a speed of 12 m/s. The solution contains 2.0 mol $\mathrm{dm^{-3}}$ of sodium chloride, so 250 $\mathrm{cm^3}$ of it holds 0.5 mol. Near the surface of the earth, free fall has an acceleration of 9.81 m $\mathrm{s^{-2}}$. The density of ice is 0.92 $\mathrm{g/cm^3}$, so a block of volume 60 $\mathrm{cm^3}$ has a mass of 55 g.

The intensity of sunlight at the ground is roughly 1020 W $\mathrm{m^{-2}}$ on a clear day. The reservoir holds 6.0 $\mathrm{km^3}$ of water spread over 150 $\mathrm{km^2}$. A steel wire with a cross-section of 2.0 $\mathrm{mm^2}$ carries a load of 900 N, which is a stress of 450 $\mathrm{N/mm^2}$. At sea level the density of air is about 1.2 kg $\mathrm{m^{-3}}$, and a barometer reading corresponds to a pressure of 2.5 times 10^5 $\mathrm{N/m^2}$.

The specific heat capacity of copper is 0.39 J $\mathrm{g^{-1}}$ $\mathrm{K^{-1}}$. At sea level the density of air is about 1.2 kg $\mathrm{m^{-3}}$, and a barometer reading corresponds to a pressure of 1.2 times 10^5 $\mathrm{N/m^2}$. Near the surface of the earth, free fall has an acceleration of 9.8 m $\mathrm{s^{-2}}$. The solution contains 0.1 mol $\mathrm{dm^{-3}}$ of sodium chloride, so 250 $\mathrm{cm^3}$ of it holds 0.025 mol.

A steel wire with a cross-section of 8.0 $\mathrm{mm^2}$ carries a load of 900 N, which is a stress of 112.5 $\mathrm{N/mm^2}$. A cart accelerates at 2.0 $\mathrm{m/s^2}$ for 6 s and reaches a speed of 12 m/s.

Atoms and Nuclei in Running Text

The mass number of a nuclide stands before its symbol, as a raised number. The nucleus $^{14}\mathrm{C}$ decays by beta emission, and its half-life $t_{1/2}$ is 5730 years. Cobalt-60, written $^{60}\mathrm{Co}$, contains 27 protons and 33 neutrons. The isotopes $^{131}\mathrm{I}$ and $^{129}\mathrm{I}$ of iodine have the same chemical behaviour but different masses. A sample of $^{137}\mathrm{Cs}$ loses half of its atoms every 30.2 years, so the activity falls in a regular pattern.

The nucleus $^{226}\mathrm{Ra}$ decays by alpha emission, and its half-life $t_{1/2}$ is 1600 years. Polonium-210, written $^{210}\mathrm{Po}$, contains 84 protons and 126 neutrons. The isotopes $^{238}\mathrm{U}$ and $^{239}\mathrm{U}$ of uranium have